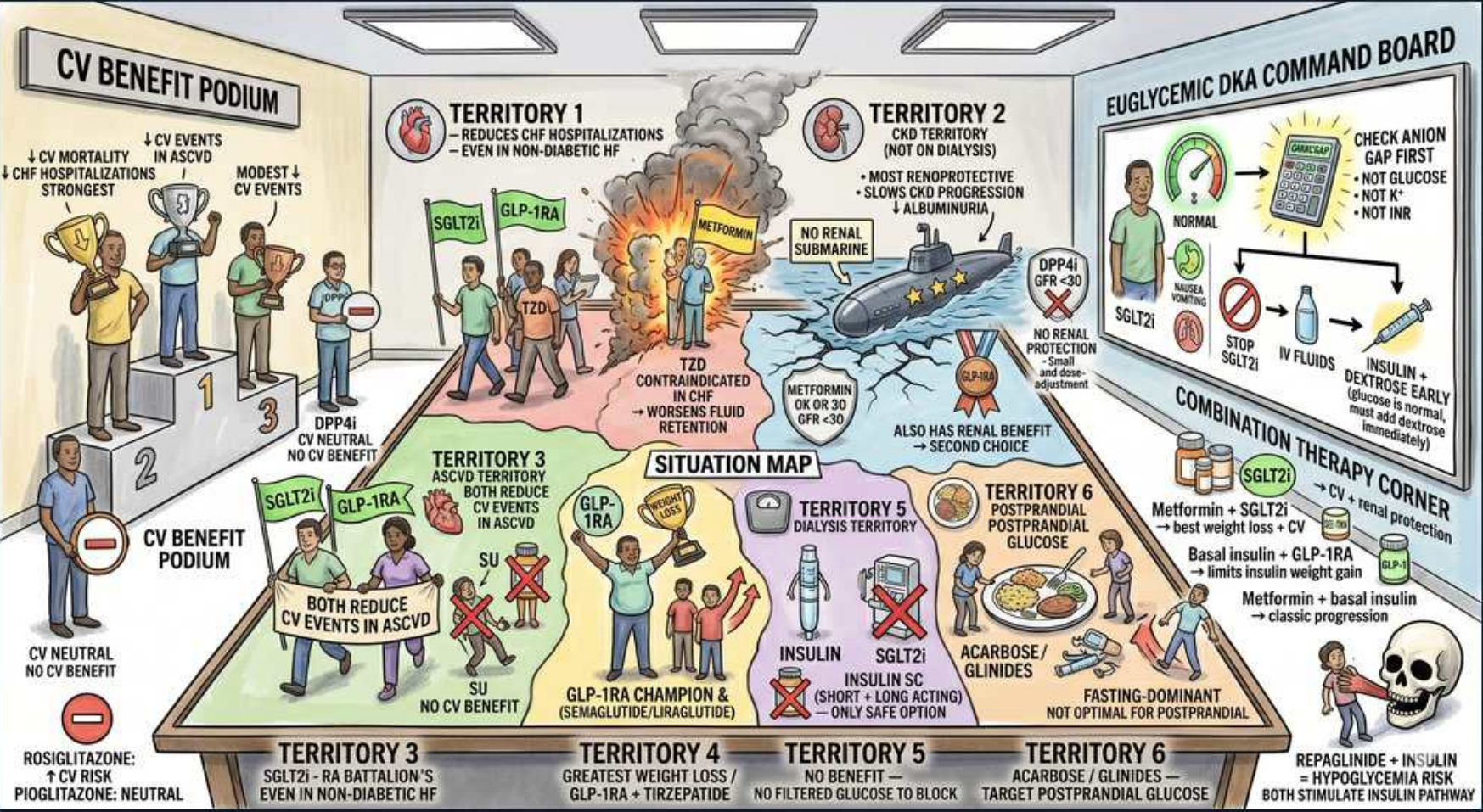


Diabetes — Emergency: DKA & HHS Resuscitation



1. DKA precipitants

WHAT YOU SEE

'Precipitants': infection, missed insulin, MI, stroke, SGLT2i.

WHAT IT ENCODES

Always hunt the trigger — DKA rarely happens without one. The 5 I's: Infection (most common — pneumonia, UTI; check WBC, cultures, CXR), Infarction (MI — often silent in diabetics; get ECG/troponin), Insulin missed/nonadherence (leading cause in young T1DM), Ischemia/stroke, and Iatrogenic/Intoxication (SGLT2i, steroids, antipsychotics, cocaine). A modestly elevated WBC is expected from stress/dehydration, but bandemia or fever points to true infection.

BOARD TRAP

WRONG: 'Treat the DKA and discharge once glucose normalizes without searching for a cause.' Missing the precipitant (e.g. silent MI or pyelonephritis) is a classic error. Discriminator: an MI or sepsis can both cause AND complicate DKA — the underlying trigger must be identified and treated.

2. Step 1 — Fluids first

WHAT YOU SEE

Step 1: 0.9% NS first, rapid; NOT insulin first.

WHAT IT ENCODES

FIRST step in DKA = aggressive IV fluids, isotonic 0.9% NS, ~1-1.5 L in the first hour (15-20 mL/kg), NOT insulin. Patients have a 3-6 L deficit; fluids alone drop glucose by hemodilution + improved renal perfusion/clearance. Volume expansion also restores tissue perfusion before insulin shifts K⁺ and water intracellularly.

BOARD TRAP

WRONG: 'Bolus IV insulin as the first intervention.' Giving insulin before volume resuscitation worsens hypovolemic shock (intracellular fluid shift) and can precipitate fatal hypokalemia. Discriminator: fluids first, ALWAYS — and check K⁺ before the insulin drip.

3. Step 2 — Insulin infusion

WHAT YOU SEE

Step 2: 0.1 U/kg/hr infusion (+/- 0.1 U/kg bolus); pause button.

WHAT IT ENCODES

After fluids are started and K⁺ confirmed ≥ 3.3 , begin regular insulin IV infusion at 0.1 U/kg/hr (optionally a 0.1 U/kg bolus first; bolus is omitted in children). Target glucose fall ~50-100 mg/dL/hr; if it does not drop ~10% in the first hour, increase the rate or recheck fluids. Insulin halts ketogenesis and lipolysis — this, not glucose lowering, is the therapeutic goal.

BOARD TRAP

WRONG: 'Hold insulin entirely once the drip is started until labs return.' OR 'Give a large SC insulin dose.' The 'pause button' applies only when K⁺ < 3.3 . Discriminator: in DKA, IV infusion (not SC) is standard, and the only reason to hold it is K⁺ < 3.3 .

4. DKA clinical picture

WHAT YOU SEE

Deep rapid Kussmaul breathing, acetone breath, nausea/vomiting, abdominal pain.

WHAT IT ENCODES

DKA presentation: Kussmaul respirations (deep, rapid breathing = respiratory compensation for metabolic acidosis, blowing off CO₂), fruity/acetone breath (exhaled acetone), nausea, vomiting, and diffuse abdominal pain that can mimic an acute abdomen. Onset is rapid (< 24 h), typically T1DM/younger. The abdominal pain correlates with the degree of acidosis and resolves as the gap closes.

BOARD TRAP

WRONG: 'The abdominal pain mandates surgical exploration / CT for appendicitis.' DKA-related abdominal pain usually resolves with correction of acidosis — treat the DKA first, reassess. Discriminator: pain that clears as the anion gap closes was metabolic, not surgical.

5. K⁺ management panel

WHAT YOU SEE

K⁺ < 3.3 stop insulin; K⁺ 3.3-5.0 add K⁺ to fluids; K⁺ > 5.0 start insulin, monitor.

WHAT IT ENCODES

Total-body potassium is profoundly DEPLETED in DKA (osmotic diuresis + acidosis shifting K⁺ out of cells), even when serum K⁺ reads high. Rules: K⁺ $< 3.3 \rightarrow$ HOLD insulin, replete K⁺ first (insulin would crash it fatally). K⁺ 3.3-5.0 $\rightarrow$ give insulin AND add 20-30 mEq K⁺ per liter. K⁺ $> 5.0 \rightarrow$ start insulin, no K⁺ yet, recheck q2h. Insulin and correction of acidosis both drive K⁺ intracellularly.

BOARD TRAP

WRONG: 'Serum K⁺ is 5.5, so the patient has too much potassium — withhold replacement entirely and don't worry about it.' Total-body K⁺ is low; once insulin/fluids start, serum K⁺ plummets. Discriminator: a 'normal' or high serum K⁺ in DKA still means depletion — anticipate the fall.

6. Bicarbonate — only if pH <7.0

WHAT YOU SEE

Bicarb only if pH < 7.0 ; NOT routine; not for pH > 7.0 .

WHAT IT ENCODES

Bicarbonate is NOT routine in DKA — insulin + fluids correct the acidosis by clearing ketones. Reserve it for severe acidemia pH < 7.0 (some say < 6.9), where myocardial contractility/pressor response is impaired. Harms: paradoxical CSF acidosis, worsened hypokalemia, delayed ketone clearance, rebound alkalosis, and possible cerebral edema in children.

BOARD TRAP

WRONG: 'Give an amp of bicarb for a pH of 7.1 to speed recovery.' At pH > 7.0 bicarb provides no benefit and adds risk. Discriminator: the only DKA bicarb indication is pH < 7.0 — moderate acidosis is treated with insulin and fluids alone.

7. HHS picture

WHAT YOU SEE

'HHS Resuscitation': severe AMS, glucose extremely high, minimal ketones, hyperosmolar.

WHAT IT ENCODES

HHS = Hyperosmolar Hyperglycemic State: profound hyperglycemia (typically >600 mg/dL), serum effective osmolality >320 mOsm/kg, severe altered mental status, with little/no ketosis and pH >7.3, HCO₃ >18. Occurs in T2DM/elderly because residual insulin suppresses ketogenesis but not hyperglycemia. Onset is insidious (days). Mortality (~10-20%) is HIGHER than DKA, mostly from the precipitant and the patient's age/comorbidities.

BOARD TRAP

WRONG: 'The patient is obtunded with glucose 800 but pH 7.35 and no ketones — this can't be a hyperglycemic emergency.' That IS HHS. Discriminator: HHS is defined by osmolality + AMS, NOT by acidosis/ketones; degree of AMS tracks osmolality (>320).

8. HHS dehydration / sunken eyes

WHAT YOU SEE

Sunken eyes, dry mucous membranes, hypotension, tachycardia.

WHAT IT ENCODES

HHS produces extreme dehydration — free-water deficit ~8-12 L, larger than DKA's ~3-6 L — because the osmotic diuresis runs for days. Signs: sunken eyes, dry mucous membranes, poor skin turgor, hypotension, tachycardia, and AMS proportional to hyperosmolality. Calculate effective osm = 2×Na + glucose/18 (urea is freely permeable, so it's excluded from the 'effective' figure).

BOARD TRAP

WRONG: 'AMS in this elderly diabetic must be a primary CNS event (stroke), so go straight to neuro workup.' Hyperosmolar dehydration alone causes the obtundation, which reverses with fluids. Discriminator: check glucose/osm before attributing AMS to a stroke — though a precipitating stroke must still be excluded.

9. HHS fluid treatment

WHAT YOU SEE

HHS fluid: total deficit 8-12 L (more than DKA), 0.9% NS first, then 0.45% NaCl by Na/osm.

WHAT IT ENCODES

HHS treatment is fluid-driven: start 0.9% NS (1-1.5 L/hr initially) to restore perfusion, then switch to 0.45% NaCl guided by corrected sodium/osmolality (if corrected Na is high/normal → half-NS; if low → stay on NS). Insulin is also given (0.1 U/kg/hr) but is secondary — fluids do most of the glucose lowering. Aim to drop osmolality no faster than ~3 mOsm/kg/hr.

BOARD TRAP

WRONG: 'Correct the glucose/osmolality rapidly to normal in the first few hours.' Over-rapid osmolar correction risks cerebral edema and central pontine injury. Discriminator: lower osmolality SLOWLY; fluids (not aggressive insulin) are the mainstay in HHS.

10. Euglycemic DKA corner

WHAT YOU SEE

'Euglycemic DKA Corner': SGLT2i; glucose normal but in DKA; don't be fooled.

WHAT IT ENCODES

Euglycemic DKA = full ketoacidosis (high anion gap, ketones, low pH) with glucose <250 (often 150-200). Causes: SGLT2 inhibitors (glucosuria masks hyperglycemia), pregnancy, prolonged fasting/starvation, heavy alcohol use, and reduced carbohydrate intake. Treat exactly like DKA but add dextrose EARLIER (D5/D10) so you can keep insulin running to clear ketones without causing hypoglycemia.

BOARD TRAP

WRONG: 'Glucose is 180, so rule out DKA and stop the workup.' SGLT2i-associated euglycemic DKA is easily missed this way. Discriminator: in any acidotic patient on an SGLT2i, check the anion gap and serum/urine ketones regardless of a near-normal glucose.

11. Anion gap drives diagnosis

WHAT YOU SEE

'Check anion gap first' — not glucose, not K⁺, not Na⁺.

WHAT IT ENCODES

The anion gap (Na – [Cl + HCO₃], normal ~8-12) is BOTH the diagnostic marker and the resolution marker for DKA. Beta-hydroxybutyrate is the dominant ketone (measure it directly; the nitroprusside/urine test detects acetoacetate and underestimates early severity). Resolution = anion gap normalized (and HCO₃ ≥15-18, pH >7.3), NOT a normal glucose.

BOARD TRAP

WRONG: 'Follow the urine ketone dipstick to gauge improvement.' As DKA improves, beta-OHB converts to acetoacetate, so urine ketones can paradoxically RISE while the patient is getting better. Discriminator: track the anion gap (and serum beta-OHB), not urine ketones, for resolution.

12. Add dextrose at glucose ~200

WHAT YOU SEE

When glucose ~200-250, add dextrose, keep insulin running.

WHAT IT ENCODES

Once glucose falls to ~200 mg/dL (DKA) or ~250-300 (HHS), add D5 (or D5-half-NS) to the IV fluids so you can CONTINUE the insulin infusion. Insulin must keep running to suppress ketogenesis and close the anion gap; dextrose just prevents hypoglycemia in the meantime. You can also titrate the insulin rate down slightly while keeping it on.

BOARD TRAP

WRONG: 'Glucose hit 200, so turn off the insulin drip.' Stopping insulin halts ketone clearance and the gap reopens (rebound ketoacidosis). Discriminator: at glucose ~200, ADD dextrose and KEEP insulin — never stop insulin to chase a normal glucose.

13. Inpatient management — no sliding scale

WHAT YOU SEE

'Inpatient Management Station': target 140-180; <110 wrong; NICE-SUGAR (tight = worse); never sliding scale alone; basal-bolus.

WHAT IT ENCODES

Non-critically-ill inpatient glucose target is 140-180 mg/dL (per NICE-SUGAR, tight control <110 increased mortality from hypoglycemia). Use scheduled basal-bolus (basal + nutritional + correction) insulin, NOT sliding scale alone. Sliding scale is purely reactive — it chases highs after they occur and never provides basal coverage, causing wide swings.

BOARD TRAP

WRONG: 'Manage the hospitalized diabetic with sliding-scale insulin only' OR 'aim for tight control <110.' Both worsen outcomes. Discriminator: basal-bolus with a 140-180 target; sliding-scale-monotherapy and tight <110 control are the wrong answers.

14. Steroid-induced DM corner

WHAT YOU SEE

'Steroid DM Corner': NPH morning insulin; steroid raises postprandial/daytime glucose; FPG often normal.

WHAT IT ENCODES

Glucocorticoids cause predominantly POSTPRANDIAL and afternoon/evening hyperglycemia with a relatively spared morning fasting glucose, because once-daily morning prednisone peaks midday. NPH insulin given WITH the morning steroid dose matches this pharmacokinetic curve (NPH peaks ~6-8h later, in the afternoon). Monitor pre-lunch/pre-dinner and bedtime glucose to catch the pattern.

BOARD TRAP

WRONG: 'Fasting glucose is normal, so the steroids aren't causing hyperglycemia.' Steroid hyperglycemia characteristically spares the morning FPG. Discriminator: a normal FPG does NOT exclude steroid-induced DM — you must check afternoon/postprandial values.
